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Review

Analytical approaches to differential extraction for sexual assault evidence

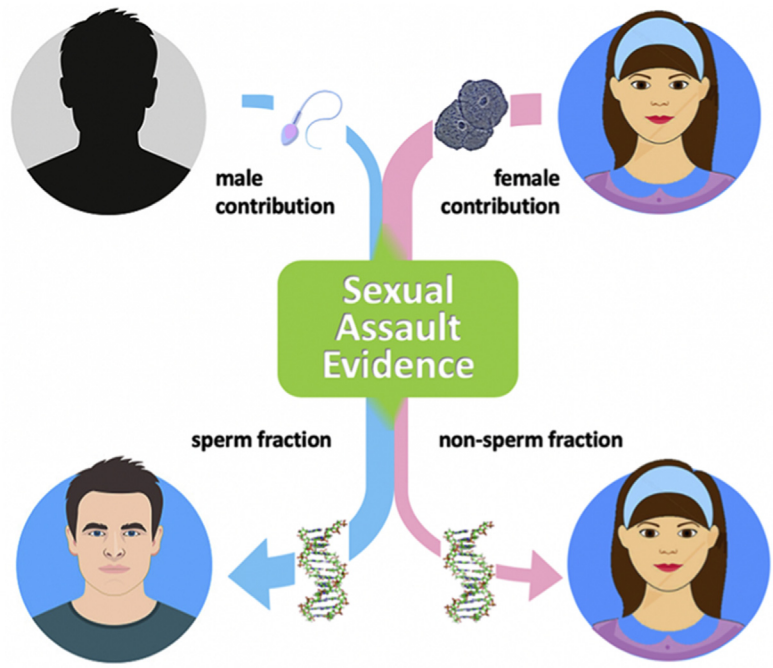
Charles Clark [a, 1], Rachelle Turiello [a, 1, *], Robin Cotton [d], James P. Landers [a, b, c]

[a] Department of Chemistry, University of Virginia, Charlottesville, VA, United States
[b] Department of Forensic Science, Boston University, Boston, MA, United States
[c] Department of Mechanical and Aerospace Engineering, University of Virginia, Charlottesville, VA, United States
[d] Department of Pathology, University of Virginia Health Science Center, Charlottesville, VA, United States

HIGHLIGHTS

- Sexual Assault and Evidence Collection Kits often contain a mixture of cell types.
- Differential Extraction (DE) enriches part of the sample with sperm cell and the other with non-sperm cells.
- Conventional DE uses chemical lysis, manual centrifugation, and washing.
- Several alternative methods exist, which eliminate many labor-intensive steps.
- Analytical requirements of DE include isolation of male DNA and sperm cell recovery.

GRAPHICAL ABSTRACT



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ABSTRACT

Many forensic laboratories face growing demands for the processing of DNA evidence from sexual assault investigations. In these cases, evidence collected from the crime scene or from the victim in the form of a Sexual Assault and Evidence Collection Kit (SAECK) typically contains a mixture of cells from at least two donors. Isolation of DNA contributions to link a sample to an alleged offender requires precise chemical treatment of each sample with the goal of separating epithelial cells from non-sperm cells. Currently, the

* Corresponding author.
E-mail addresses: rat3a@virginia.edu, rachelleturiello@gmail.com (R. Turiello).
[1] Both contributed equally to this review.

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Forensic chemistry
Sexual assault and evidence collection kit
Selective degradation

vast majority of laboratories employ differential chemical lysis methods that require lengthy incubations and several manual steps, preventing complete automation. Numerous alternative methods for the differential extraction (DE) of sexual assault evidence have been developed to provide a solution to the growing backlog of samples observed in the US and other countries. Here, we will discuss the predominant methodology for the DE of DNA from sexual assault samples and review alternative approaches from literature. We illustrate three criteria that provide a measure of success in performing these types of chemical separations and examine all methods based upon these expectations. We conclude by providing some general insight into the application of DE techniques in forensic laboratories and discuss the potential future directions of alternative technologies.

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1. Introduction

Forensic laboratories face persisting demands for processing DNA evidence from sexual assault investigations, which may exceed current throughput capabilities. Sexual assault samples collected from bodily fluids, tissues, clothing, and various other materials, typically contain mixtures of cells from both the alleged perpetrator and victim. Additionally, DNA from evidence samples may be degraded or present in low concentrations. All of these factors increase the difficulty of obtaining a DNA profile amenable to individualization from a given contributor. The best and most common approach in analysis of sexual assault samples is to separate sperm from all other cell types present in the sample. Most current protocols for sperm cell isolation rely on differential chemical lysis of victim epithelial cells, followed by manual centrifugation and washing steps to separate the DNA from non-sperm cells, mainly from female origin, from any sperm cells. This process is lengthy, labor-intensive, and difficult to automate. Several alternatives, however, have been developed in recent years, and may assist with the processing of sexual assault samples. This review highlights a number of innovations focused on improving or complementing differential extraction (DE); the process that separates spermatozoa from non-sperm cells in forensic mixtures. This review is not comprehensive, but instead focuses on what we believe are key developments exploiting the benefits that accompany unique chemistries and physics in the microscale regime.

1.1. Sexual assault in the United States and worldwide

Global estimates of the prevalence of sexual assault are difficult to quantify or compare, as representative data is influenced by a dynamic spectrum of factors, including conflicting definitions of rape, divergent cultural contexts, gender-based attitudes, and sexual preferences [1]. Defining metrics is further complicated by the broad range of sexually violent acts, and dissimilar regional interpretations of consent and coercion [2,3]. Existing data represent only those sexual assault incidents compiled from police records, clinical settings, non-governmental organizations and survey research [3]. Fig. 1 provides illustration on the collation of statistics from multiple international studies in terms of the relationship between the circumstances surrounding sexual violence and its reporting (reprinted with permission [4]). Largely, analyses suggest that the majority of sexual assaults go unreported [1,5,6].

Although the occurrence of rape is vastly underreported in official statistics [7,8], the number of documented assaults is still disturbingly common. The World Health Organization (WHO) estimates that nearly one in three women (35%) worldwide have experienced sexual violence in their lifetime [9]. At present, South Africa has the highest rate of rape in the world, with approximately

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132 documented incidents per 100,000 people. However, the self-reported 27.6% male offender rate in South Africa indicates there are many more unreported assaults and rapes, according to a study across multiple provinces in the region [10,11]. Several adjacent countries, including Botswana, Lesotho, and Swaziland, also experience similarly high occurrences of rape according to recent statistics [10]. Comparable reporting from six countries across Asia and the Pacific estimate that nearly a quarter of the 10,000 men interviewed admitted to perpetrating rape against a woman or girl [12]. In the United States, there are over 300,000 victims of sexual assault each year [13], with nearly one in five women claiming to be the victim of rape [14]. Other parts of North America share these troubling figures; for example, the Canadian Center for Justice Statistics reports roughly 22 cases of sexual assault per 1,000 residents, although it is noteworthy that this data is self-reported [15]. Similar numbers of sexual violence may be found in the European Union, with a particularly high prevalence of rape occurring in Nordic countries [16], and a general trend suggesting that rape reporting is significantly higher in the UK and Scandinavian countries when compared with Central and Eastern Europe [17]. A likely reality is that occurrences of sexual assault and rape are high everywhere, and with a lower probability that a specific isolated case will be investigated by law enforcement personnel and successfully prosecuted.

Identification of an offender is vital for sexual assault investigations, which rely on the timely reporting of cases, examination of the victim, and the collection and analysis of physical evidence. Unfortunately, sexual violence is vastly under-reported, and despite the introduction of laws to define and criminalize sexual assault and rape, the rates of evidence analysis and conviction in both developed and less-developed countries remain low [18]. When these cases are reported to law enforcement and resources are available, corroborating evidence may be collected from the crime scene, potential suspects, or from the victim [19]. Samples can be collected from a victim's body during physical and gynecological examination by taking swabs (vaginal, cervical, external genitalia, anal, etc.). Additionally, clothes worn by the victim at the time of the incident should be carefully examined for any biological material or stains and properly collected by a Forensic Medical Practitioner (FMP) or gynecologist [20,21]. In many countries, the development of a standardized method for the retrieval of intimate evidence exists in the form of a Sexual Assault and Evidence Collection Kit (SAECK) [21]. SAECKs, which are often referred to as simply Sexual Assault Kits (SAKs) or *rape kits*, may vary in their contents by country and even between jurisdictions, but usually include collection instructions, sampling swabs and containers, and documentation forms [18]. One common method for detection of an alleged perpetrator of a crime of sexual violence is to recover sperm cells from the evidence contained in the SAECK.

1.2. Analysis and backlog of sexual assault evidence

Forensic laboratories across the United States are tasked with processing hundreds of thousands of pieces of biological evidence from crimes each year, and approximately three quarters of these samples require DNA analysis for investigation [22]. This involves taking samples of interest through 'sample preparation' and genetic analysis. The 'sample preparation' involves a series of analytical steps that isolate the DNA from the sample and remove inhibitors that can affect DNA amplification. Once PCR-ready DNA is attained, it undergoes thermocycling-based amplification via PCR with a multiplexed primer panel that directs the amplification of hundreds of millions of copies of short tandem repeat (STR) sequences in the genome with amplicons ranging in size from roughly 100–600 bases. The PCR-amplified products undergo a size-based separation by capillary electrophoresis (CE) under denaturing conditions with multicolor fluorescence detection. Software inherent to CE instruments and designed for STR analysis deconvolutes the raw multicolor (4–8 colors) fluorescence data to generate the type of STR profile given in Fig. 2, where each allele from each probed location (locus) is sized for the number of repeats [23].

A significant portion of sexual assault evidence samples are associated with cellular mixtures from multiple contributors [24–27]. These may contain several distinct cell types (sperm, epithelial, blood, bacterial, leucocytes, etc.) that vary drastically in their composition on a case-to-case basis (Fig. 3). Blood, saliva, or other biological samples mixed with sperm often cannot be directly analyzed via DNA profiling without separation of sperm and non-sperm cells. Furthermore, SAECKs may require multiple serology tests, to determine which samples should move forward for DNA analysis and where to cut the sample from large items such as clothing or bedding [28]. Due in part to the time-consuming, manual protocols required to process SAECKs, there continues to be a significant backlog of samples awaiting DNA analysis [29–33]. This backlog exceeded 150,000 samples across the United States in 2006 [30], and backlogs continue to be present in many jurisdictions; as evidenced by more recent courts in several jurisdictions [32]. Federal funding for both equipment purchases and increased staffing via grants to forensic laboratories has not been sufficient to overcome this backlog, although the throughput of laboratories has increased steadily each year since 2009 [33]. However, the demand for forensic DNA casework has increased at an even higher rate, compounding the problem [33]. The societal impact of this backlog

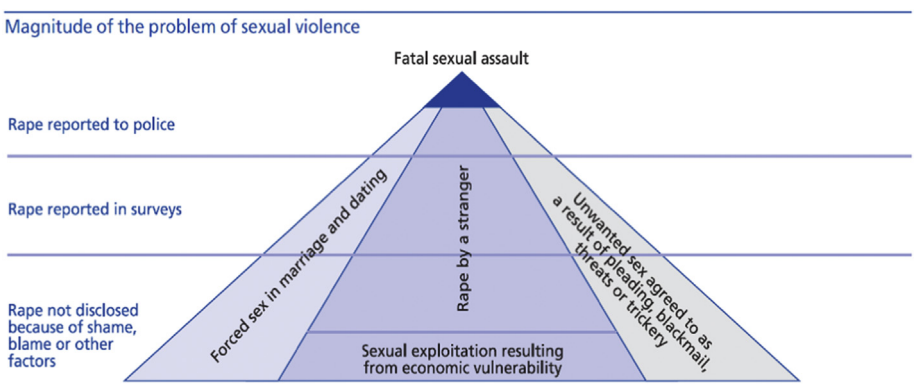


Fig. 1. Schematic summarizing approximate distribution of sexual violence and reporting data from international crime victim surveys conducted by the United Nations Interregional Crime and Justice Research Institute. Reprinted with permission from Ref. [4].

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is profound, as victims of sexual assault crimes may be forced to wait months, or even years before DNA evidence is available to further inform an investigation and in many cases help to identify the attacker. To address this issue, there have been several initiatives from the United States government which have increased funding and prioritized the rapid processing of sexual assault samples [34]. As recently as 2017, the National Institute of Justice committed $117 million for a 'DNA analysis and capacity enhancement program,' which was tasked with addressing the critical forensic DNA backlog [35–37]. In addition to legislation at the state and national level, programs like this have changed how sexual assault cases are reported [38] and how the evidence is collected [20,39], but not the basic methods for processing samples.

1.3. The differential extraction (DE) process

1.3.1. An ideal DE

While the purpose of this review is to overview a series of specific protocols for DE in the sections that follow, it is imperative to understand what the goal of DE is and, in an ideal world, how the end products of an "ideal DE" can be described as two-fold. First, and most importantly, an ideal DE generates a solution of male DNA, devoid of any DNA or cellular material from the victim. This is referred to as the 'sperm' or 'male fraction' which, with efficient removal of all female DNA, is ideally single-source (when only one male contributor is present). An ideal DE will also generate a 'non-sperm', or 'female fraction', containing both DNA from the victim, and DNA from the non-sperm cells of any male donor. From the sperm and non-sperm fractions, forensic analysts can proceed with DNA amplification and electrophoretic analysis to obtain DNA profiles from each fraction. An ideal differential extraction will also be fast and inexpensive. For the purposes of this review, "conventional DE" will refer to the manual sample preparation of sperm cells from materials suspected to contain mixed cell populations. The automation of DE and alternative methods described herein may incorporate either separation alone, or separation and subsequent purification.

1.3.2. Analytical requirements of DE

The challenge of executing an efficient DE is not associated with complex, instrument-dependent methods. This is primarily because the steps involved include manual processes such as detergent-based, proteinase-assisted cell lysis and centrifugation. Multiple types of DNA extraction can be used following the separation of non-sperm cell DNA from the sperm cells. The success of any particular differential extraction technique can be assessed by several criteria, and include the following:

1. **Extent of contamination of the non-sperm cell fraction with male DNA.** Contamination of the non-sperm cell fraction with male DNA is not considered to be as critical to the process of inclusion or exclusion of a suspect and the presence of male DNA in the female fraction is not unusual. Non-sperm cells present in semen such as male leucocytes or spermatocytes are subject to lysis with detergent and proteinase K as are the female epithelial cells. There may also be some lysis of sperm with the chemistry designed to lyse the female epithelial cells. Since the elapsed time between an assault and sample collection is variable, as are the conditions of sample storage post collection, the sperm cell wall or sperm DNA may accumulate damage resulting in some premature sperm cell lysis into the non-sperm fraction. While the presence of some male DNA in the non-sperm fraction is common, unintended loss of sperm DNA to the non-sperm fraction is detrimental in that it reduces the amount of DNA available for analysis in the sperm cell fraction.
2. **Overall recovery of sperm cell DNA.** Measurement of sperm DNA recovery should be done during validation of a differential extraction process using known samples as it is not possible to obtain this information later from evidence samples. The sperm DNA yield can be broken down into loss of sperm/DNA in the differential cell separation portion of the process and then loss of DNA in the DNA purification steps. Discrete recovery yields are difficult to obtain in mixed samples. However, in process development, there must be some measure of the efficiency of recovery of sperm cells from the original samples. This could include consideration of the extent of release of sperm cells from the original substrate, loss of sperm with each centrifugation (or other type of step), and DNA loss in the DNA extraction process which follows lysis of sperm in the sperm pellet.
3. **The ratio of male to female DNA in the final sperm cell extract.** The ultimate goal of a successful differential procedure is to obtain a single-source male DNA profile from a previously mixed cell sample in which the victim would likely be the major contributor to the unseparated mixture. Without proper cell separation to isolate the sperm cells, DNA from the male contribution may be minimal in comparison to female DNA and thus more difficult to statistically quantify. Ratios of approximately 10:1 or greater will effectively produce a single source male DNA profile, such that mixture interpretation is not necessary. In some cases, a mixture ratio of approximately 4:1 may be sufficient to conclusively isolate a major male contributor from a profile. Lower ratio scenarios may be further informed by the availability of a known victim standard and/or the use of probabilistic genotyping software.

1.3.3. Conventional DE

The foundational paper on conventional DE was published in 1985 by Gill et al., and describes their approach for separating sperm cell nuclei from vaginal cellular debris, leading to a 'DNA fingerprint' that can aid in identifying a suspect [40]. Using this method, sexual assault samples (typically a cotton swab or cutting from an article of clothing) are subjected to a preferential lysis that breaks open all non-sperm cells and releases their genetic material, leaving sperm cells intact. Following Gill et al., the initial digestion of epithelial cells is performed with proteinase K (Pro-K) and sodium dodecyl sulfate (SDS), leaving sperm cells intact. Adding the modification published by Yoshida et al., the lysate is centrifuged and washed with the initial lysis solution prior to recovering the sperm cell pellet [41]. The sperm cells are then lysed, typically using a surfactant, Pro-K, and dithiothreitol (DTT) [40], resulting in a purified sperm fraction which ideally contains only DNA from the assailant and/or consensual partner. This process is illustrated in Fig. 4 showing the steps required to isolate sperm cells via centrifugation. Once male DNA is isolated in the male fraction, the forensic analyst can continue to quantification, amplification, and separation of the DNA, via qPCR, PCR and capillary electrophoresis, respectively. Forensic laboratories typically generate high quality DNA profiles from the conventional DE approach and the obtained DNA profiles can provide valuable evidence for criminal investigations.

1.4. Shortcomings of conventional DE

It is important to note that for conventional DE, the centrifugation and washing steps each require five minutes or longer, and are repeated up to five times before a clean male fraction can be obtained [42]. The number of wash repetitions used differs between laboratories and the number needed cannot easily be known in advance with traditional microscopy of the sample. This process

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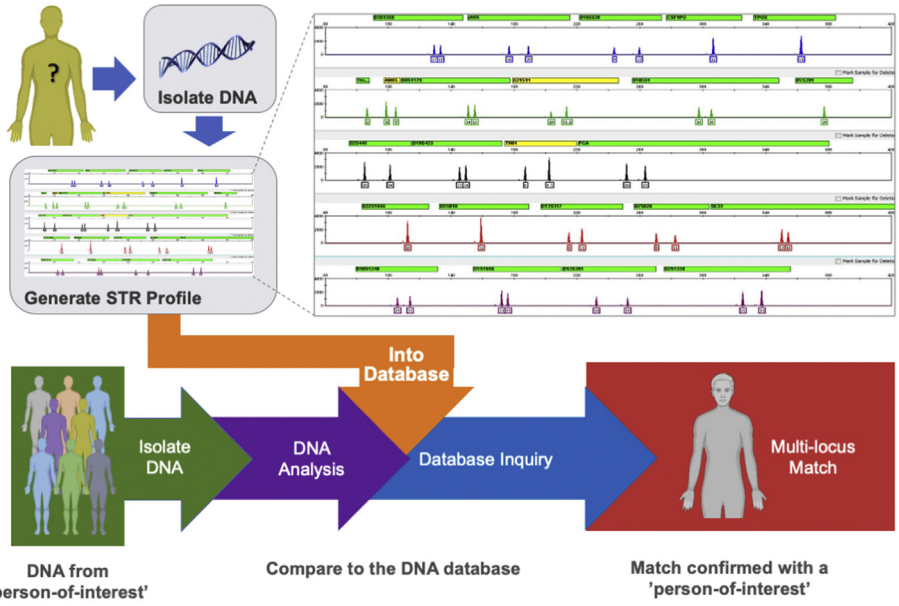


Fig. 2. Short tandem repeat profiling in human identification. When DNA is isolated from a single person, it can be amplified, separated based on fragment size, and analyzed as part of identifying an individual.

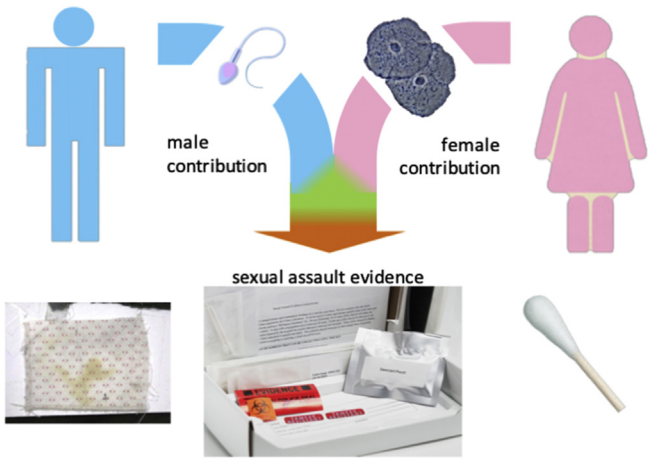


Fig. 3. The complexity of sexual assault evidentiary material and the need for differential extraction processing.

is generally effective at isolating sperm cells, but it is time-consuming (potentially several hours per sample) and requires a skilled forensic analyst. The quality of the separation can vary drastically between labs and individual analysts and may be dependent on the age and condition of the evidence sample [43]. Additional differences between each laboratory process arise from variable incubation times, dependent upon the specifics of the DE method used. There are various commercial options for DE kits and reagents, however, these methods still involve multiple manual washing steps, and some require additional DNA concentration after DNA isolation. (Table 1, adapted from Vuichard et al. [43]). Some of the time constraints associated with sample preparation and downstream processing are unavoidable, as cellular lysis, DNA quantification, DNA amplification, and DNA separation will remain constant regardless of the technique used to capture sperm cells. However, the hands-on steps of washing the sperm pellet and subsequent centrifugation are most problematic, as they are difficult to automate and expose the sample to potential contamination with each successive treatment. For this reason, the most promising research to improve SAECK processing involves exploring alternative methods to more effectively capture sperm cells, while reducing the analyst time and contamination potential for each sample.

2. Automation of conventional DE

In response to the demand on forensic laboratories, several commercialized methods have been developed for automated DE. However, due to the complex liquid handling and sample transfer steps necessary, a common approach is to automate select portions of the DE process. One commonly used system is the AutoMate Express™ Forensic DNA Extraction System (Applied Biosystems), which was developed to automatically isolate DNA from a variety of forensic samples. Liu et al. demonstrated automated extraction of male DNA from mock sexual assault samples using the AutoMate Express™, after first performing a manual DE in accordance with the Gill protocol [40,44]. The automated extraction implemented PrepFiler™ (Applied Biosystems) reagents, and showed sufficient quantities of DNA recovered from both the sperm and non-sperm fractions, as well as removal of PCR inhibitors during DNA extraction. Like the AutoMate Express™, the BioRobot EZ1 (QIAGEN) performs automated extraction of DNA from a variety of forensic substrates. EZ1 is one of the technologies that relies on magnetic beads, and has been shown to produce similar DNA yields to organic extractions, while also removing PCR inhibitors [45]. In a head-to-head comparison, using mock forensic samples, the EZ1 demonstrated the highest DNA yield for many sample types but for semen on cloth, the Automate express and the Maxwell outperformed the EZ1 in this limited study [46]. However, while these systems do reduce the number of manual steps in a DE, they still rely on centrifugation for the physical isolation of sperm cells. They offer controlled, reproducible lysis of sperm heads for extraction and purification of male DNA, but do not, and cannot, replace the separation step needed to isolate sperm cells from a sexual assault sample.

Integrating the isolation and purification workflow required for SAECKs into a single automated workstation is a significant

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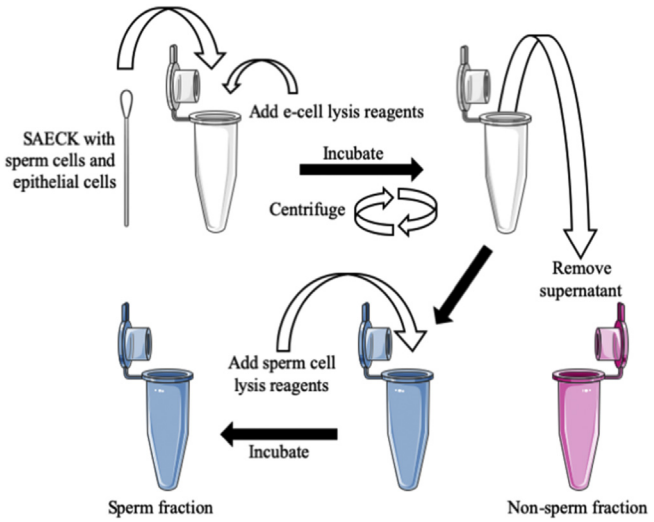


Fig. 4. Schematic of a traditional differential extraction protocol. A cutting of the sample containing sperm cells and epithelial cells is combined with epithelial cell lysis reagents, liberating non-sperm DNA. Using centrifugation, sperm cells are pelleted to the bottom of the tube, and the supernatant is removed as the non-sperm fraction. Sperm cell lysis reagents are added to the pellet of sperm cells and incubated, breaking them open and liberating male DNA, to generate the sperm fraction.

challenge in forensic science. One commercial system for multiplexed DE is the QIAcube, which contains a centrifuge, heated orbital shaker, and robotic liquid handling. Samples undergo differential lysis, followed by sperm cell centrifugation, washing, and lysis, which are performed with some user intervention. Subsequent DNA purification, quantitation, and amplification are performed separately, and pairing the QIAcube (QIAGEN) with the EZ1 and/or Rotor-Gene Q can further minimize the hands-on steps required for each sample. With this workflow, the equipment cost increases and the process is still semi-automated rather than truly automated. The QIAcube has been evaluated in comparison with other technologies, including conventional DE, for its ability to isolate and purify sperm cells from SAECKs. Schwerdtner et al. found that the QIAcube recovered 58% of the male DNA present in a sample, and improved the female-to-male DNA ratio by 188X compared to the un-separated mixture [47]. An additional study compared performance of the automated QIAcube protocol to a manual DE kit, as well as a traditional organic extraction. Those findings showed no significant difference in male DNA yield from the sperm fractions generated by each method, indicating that the QIAcube can isolate sperm cells as effectively as traditional techniques when a large number are present in the starting sample [48]. The performance of the QIAcube system suggests that, for laboratories who can afford the initial instrument cost, this automated method is a viable alternative to conventional DE.

Several recent studies from Timken et al. address improvements to the differential extraction method and report on the performance of another commercial system, the AutoLys STAR (Hamilton) liquid handling workstation [49,50]. Like the QIAcube, the AutoLys incorporates sample incubation, shaking, and centrifugation steps in an automated fashion, processing up to 24 samples in a hands-free workflow. Implementing a proprietary AutoLys-A tube and carrier, the system performs multiple washing and lysis steps to remove non-sperm cells DNA post lysis. The AutoLys system was evaluated with mock sexual assault samples, which were prepared with buccal and vaginal swabs, then spiked with varying dilutions of semen ranging from 0.2 μl to 0.02μl and results were compared to use of conventional DE. These amounts of starting semen are considerably below the 3 μl used by Goldstein et al. The STR results from the AutoLys samples and the conventional DE were almost identical by yield of male DNA and also by assessing the amount of female DNA remaining in the sperm fraction. More importantly, the sample numbers reported are small and the final ratio of male to female DNA in the sperm fraction for samples using 0.1 and 0.02μl semen ranges from approximately 3:1 to 1:2. While not currently prepared to replace conventional DE for all sample types, this automated system could significantly enhance throughput for some laboratories.

Another technique seeking to provide automated DE is the Differex™ System (Promega) [51]. This approach combines differential centrifugation with phase separation, while still employing the differential lysis and washing steps of a conventional DE. Differex™ is compatible with an array of robotic workstations (Biomek 3000, Freedom EVO, etc.), which allow for automated sample handling. Following proteinase K epithelial cell digestions, sperm cell isolation is achieved using a separation solution that has a density greater than water, but less dense than sperm cells. This forms a barrier between the sperm cell pellet and aqueous solution of free DNA, which can be easily removed for the non-sperm fraction. For sperm cell lysis and isolation of male DNA, DNA IQ™ Resin (Promega) is added to the sperm fraction, which 'caps' the sperm cell pellet and eliminates the need for additional centrifugation after washing steps. Differex™ has been evaluated in comparison to the conventional DE protocol, and successfully extracted male DNA from mixed stains with similar efficiency as the conventional approach [52]. Another laboratory, however, evaluated the Differex™ system and chose not to implement it permanently, due to incompatibility with downstream microscopy for sexual assault samples [53]. It is important to note, however, that different labs can choose to implement microscopy before or after certain steps of DE, and Differex™ may be suitable in some circumstances. The Differex™ system requires slight modifications to become fully automated, and may not be suited for all laboratories, but the innovative combination of phase separation and differential centrifugation is an intriguing alternative to conventional DE.

3. Alternative approaches to conventional DE

The conventional approach for DE can best be summarized as functional, but imperfect, i.e., it accomplishes the task but with clear-cut drawbacks. Several academic research groups and companies have addressed improving DE, and these efforts have produced some noteworthy developments and new technologies. Some of the most promising approaches are detailed here.

Table 1
Differential extraction protocols employed in eight European forensic laboratories for processing sexual assault samples. Reprinted with permission from Ref. [43].

Initial lysis	Number of washes	Secondary lysis	DNA purification	Concentration instruments
Differex, pro-K	2	ATL buffer, DTT, pro-K	QIAamp DNA mini kit	Microcon
Lysis buffer, pro-K	1–4	Lysis buffer, DTT, pro-K	Organic	Precipitation/Centricon
Chelex, pro-K	3	Chelex, DTT, pro-K	Chelex	Centricon
Lysis buffer, pro-K	3	DTT, pro-K	QIAamp DNA mini kit	Centricon
ATL buffer, pro-K	3	ATL buffer, DTT, pro-K	QIAamp DNA mini kit	none

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3.1. Sperm cell capture through dielectrophoresis

The DEPArray™ (Menarini Silicon Biosystems) is a rare cell recovery system that employs dielectrophoresis to manipulate and capture individual sperm cells, providing impressive sensitivity and selectivity. DEP exploits the polarizability of cells in a nonuniform electric field, which exerts force on the cells that lead to movement. Precise manipulation of an electric field gradient allows for individual cell capture, as demonstrated by Williamson et al. [54]. Subsequent imaging of the captured cells employs a sperm-specific stain and a nucleus-specific stain, showing single cell retention (Fig. 5, adapted from Ref. [54]). Major drawbacks of this approach are: 1) that it requires more time and operator steps than most other methods, 2) it requires cell staining/fixation and lengthy incubations and, 3) as a technology developed for rare cell capture in the clinical sector, it is prohibitively expensive for routine use. An additional cost factor is the single-use, disposable (or recyclable) microfluidic cartridge, which is $300 USD. That said, it is an innovative approach that could see application with trace evidence samples having low numbers of sperm cells. Additionally, although it has not yet been demonstrated, the DEPArray™ could theoretically be used for samples where there are sperm cells from multiple assailants, or the presence of sperm cells from consensual partners in a single sample. In any of those cases, traditional DE would fail to produce a single-source male STR profile, but DEPArray™ may succeed due to its ability to capture individual sperm cells. However, the nature of the dielectrophoresis, and the fluidic control currently will not allow for the grouping of individual sperm cells based on their source.

3.2. Selective degradation of female DNA

While most methods for generating single-source male STR profiles from sexual assault samples rely only on the physical separation of the sperm cells from non-sperm material, a 'nuclease-only' approach has also been developed. This relies on a single centrifugation step to concentrate sperm cells, a protease, and detergent for epithelial cell lysis followed by use of DNase 1 enzyme to completely remove all female DNA from the sperm pellet prior to lysing the sperm cells [55–58]. Garvin et al. showed the nuclease method can generate male STR profiles of equal or higher quality than traditional DE methods (Fig. 6, from Ref. [55]).

Selective degradation has been exploited in a number of commercialized products for processing sexual assault samples, available in the Erase Sperm Isolation Kit (PTC Laboratories). This kit eliminates the washing steps of conventional DE, using selective degradation of residual non-sperm DNA to produce a clean sperm fraction. This sperm cell isolation kit has been evaluated directly with conventional DE, with mixed results. Klein et al. demonstrated that, when testing mock sexual assault samples with both methods, the Erase Sperm Isolation Kit produced sperm fractions with a higher ratio of male:female DNA (i.e., more purified), but a 6-fold lower concentration of total male DNA [59].

In an attempt to automate this approach, selective degradation of female DNA was combined with the VERSA 1100 liquid handling station (Aurora Biomed). The automated method provided similar DNA yields to the manual selective degradation approach, and, as one would expect, showed excellent reproducibility across multiple replicates [60]. An additional report found that automated selective degradation, again using the VERSA workstation, produced the same results as a conventional DE protocol, but with a 6-fold increase in speed, and without analyst intervention [61]. The effectiveness of selective degradation method (manually or robotically) makes it an important addition to traditional DNA analysis, but has not replaced the conventional DE process in a majority of forensic laboratories.

3.3. Surface modifications for sperm cell binding

While no commercial alternative to DE has supplanted the conventional method, there are many technologies in the research and development stage. An interesting approach has been developed by Inci et al. exploiting an oligosaccharide that has been shown to be crucial for sperm-egg binding during fertilization [62]. With the oligosaccharide anchored to a microfluidic chip surface, the incubation of a sample containing sperm cells and epithelial cells on-chip allow for sperm cell binding to the carbohydrate, thus retaining the sperm cells under flow as the epithelial cells are washed away (Fig. 7, reprinted with permission from Ref. [62]). This method is being developed with mock sexual assault samples, and has the distinct advantage of performing subsequent sperm cell lysis on-chip (a step not often integrated for microfluidic DE methods). This technique is currently limited to small sample volumes (5–15 μL), a relatively lengthy incubation to facilitate sperm cell capture (60 min), and has yet to be rigorously evaluated for contamination concerns. However, as the technology continues to be developed for higher throughput and efficiency and the cost, risk factors, and DNA yield are addressed, it may have the potential to supplant conventional DE with additional development.

3.4. Laser capture microdissection

Laser capture microdissection (LCM), one of the most precise capture techniques, harvests sperm cells by individually cutting them out of the substrate. As indicated by the name, this requires microscopy to identify the target cells, and a skilled user to remove them from the sample through laser manipulation [63]. As shown in Fig. 8 (reprinted with permission from Refs. [64]), individual cells of interest can be identified under magnification and removed into a microfuge tube for DNA processing. The advantage of such a methodology is clear – the ability to acquire sperm cells with (presumably) no contaminating female component. LCM was evaluated specifically for its applicability to sexual assault samples, by first staining the sample of interest to differentiate sperm cells from other cell types, harvesting sperm cells for DNA extraction, and generating short tandem repeat (STR) profiles from the purified DNA. This was highly effective for generating single-source male STR profiles, with no female DNA present after LCM [65]. Such an approach is desirable for samples that may contain low number of sperm cells. Overall, LCM can effectively separate sperm cells from mixed samples, but routinely provides sperm fractions with a low number of cells. While LCM systems have been advanced by a capability for semi-automation (e.g., the Olympus CellCut microdissection system), a major hurdle still seems to exist if the very limited adoption in the US is an indicator. This results, in part, from the high cost of the instrumentation, but the impressive selectivity is likely at the root of its successful deployment for improved forensic analysis of SAECKs [66].

3.5. Antibody-based capture methods

Several groups have reported on the feasibility of sperm cell capture using antibody-based methods, targeting biomolecules like sperm protein-10 or testicular angiotensin-converting enzyme with antibodies bound to magnetic beads [67–69]. Another antibody binding approach for sperm cell capture has been developed by Li et al. [70]. Their biological target is the motile sperm domain-containing protein 3 (MOSPD3), which is expressed on the surface of sperm cells, and can be selectively bound using monoclonal antibodies. A biotinylated MOSPD3 antibody is incubated with mixtures of sperm and epithelial cells, along with avidin-coated magnetic beads. This causes sperm cell binding to two or more

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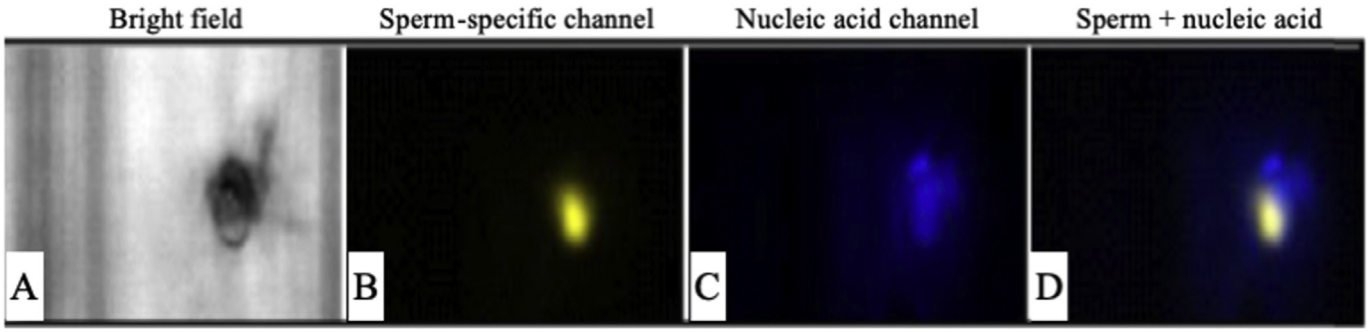


Fig. 5. Images captured from the DEPArray™ showing a bright filed image of a sperm cell (A), sperm-specific APC channel which indicates capture of a single sperm cell (B), nucleic acid DAP1 channel (C), and an overly of the sperm and nucleic channels (D). Adapted from Ref. [54].

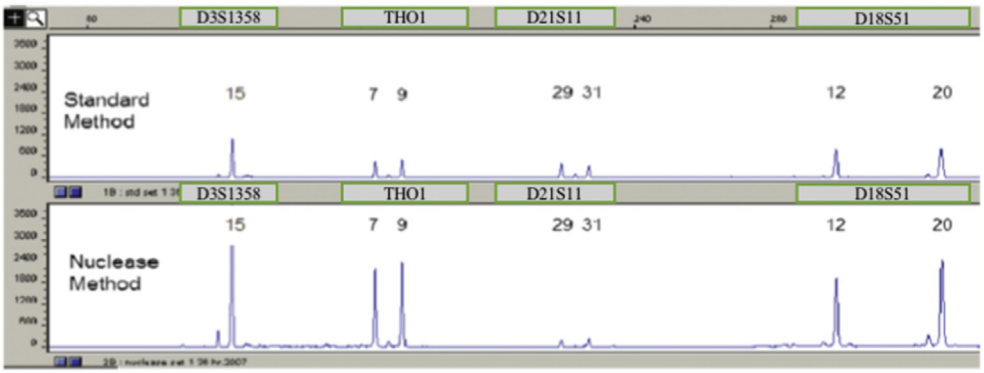


Fig. 6. Four STR loci (D3S1358, THO1, D21S11, and D18S51) from the PowerPlex16 kit for buccal swabs and male fractions from postcoital vaginal swabs. Male fractions obtained from a 36-h postcoital swab gives a concordant male profile with both the standard selective lysis protocol and the nuclease protocol. Adapted from Refs. [55].

beads, which can then be retained magnetically during washing and purification. This method was evaluated using 30 mock samples composed of equal number epithelial and sperm cells, as well as 52 authentic, non-probative SAECKs. STR results showed that, samples containing more than 10^4 sperm cells/mL, the MOSPD3 antibody was 100% successful in generating a single-source male STR profile. With samples stored for longer periods of time, however, the efficiency declined precipitously, e.g., after just 10 days, a single-source male profile could be obtained from only 16% of samples tested. This drastic reduction in capture efficiency was attributed to sperm cell degradation, and specifically, the decay of the tail, flagellum, and mid-piece of the sperm cell. That degradation resulted in fewer potential binding sites for the antibody, and thus would explain poor capture efficiency. Additionally, it is common in real-world samples, that the tail, flagellum, and mid-piece) are frequently not observed from SAECKs. A similar approach has been demonstrated by Grosjean et al., who also used spermatozoa specific antibodies coupled to magnetic beads to separate sperm from other cell types. This technology has recently been realized into a commercial product, the MACSprep™ Forensic Sperm Microbead Kit (Mitenyi Biotec), which has the potential for workflow automation.

Zhao et al. have demonstrated that, after a preferential lysis of epithelial cells, sperm cells can be immobilized using antibodies tagged to magnetic beads, which can then be removed from the sample lysate [71]. They used optical and scanning electron microscopy to show binding of intact and damaged sperm cells to the antibody-tagged beads (Fig. 9, reprinted with permission from Refs. [71]). This approach specifically targeted PH-20, a hyaluronidase located on sperm cell heads, by using anti-PH-20 immunomagnetic beads. The authors report this as the first covalently-linked sperm cell antibody capture technique, and showed single-source male STR profiles generated from mock sexual assault samples, even in extreme cases where an overwhelming number of epithelial cells were present. As with commonly used differential extraction procedures the antibody capture benefitted from the use of DNase treatment to remove residual epithelial cell DNA. While this procedure will require substantial additional validation, it shows promise and warrants further investigation.

DNA-based affinity reagents have also been used for sperm cell capture, as Katilius et al. demonstrated with an aptamer-based approach [72]. These SOMAmer (Slow Off-rate Modified Aptamers) reagents are composed of modified DNA bases, whose affinity and specificity are similar to monoclonal antibodies [73–75]. The SOMAmer reagents are immobilized on magnetic beads via biotin-streptavidin complexes, and specifically formulated to bind sperm cells on both the head and tail of the cell. The beads are incubated with the liquid sample for 30 min, allowing sperm cells to bind the aptamers, followed by a wash step to remove epithelial cells. When evaluated with mock samples, however, non-specific binding was observed between epithelial cells and the magnetic beads. This resulted in carryover of female DNA to the sperm fraction, which could lead to inconclusive STR results. An additional lysis step was necessary to mitigate this carryover, and the modified protocol produced results similar to conventional DE for some samples. As sperm cells became more dilute, however, the SOMAmer approach struggled to generate clean fractions. Despite limited applicability to dilute samples, this method may still be applicable in a forensic setting. The absence of centrifugation steps combined with the potential for automation make SOMAmers an appealing option for samples containing a large number of sperm cells, and could reduce demands on forensic analysts.

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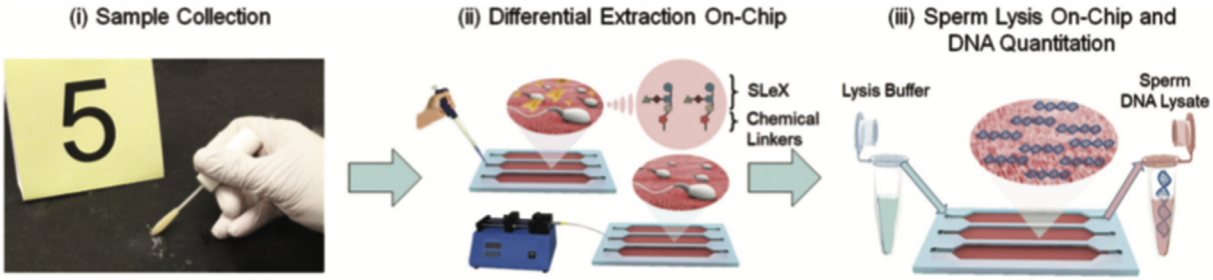


Fig. 7. After sample collection, sperm and epithelial cell mixtures are pipetted into the device and incubated for 1 h at room temperature, allowing sperm cells to bind the carbohydrate ligands. Channels are washed to remove epithelial cells, while sperm remain bound to the microdevice. On-chip lysis of sperm cells liberates the male DNA, which is collected into a tube for genomic analysis. Reprinted with permission from Ref. [62].

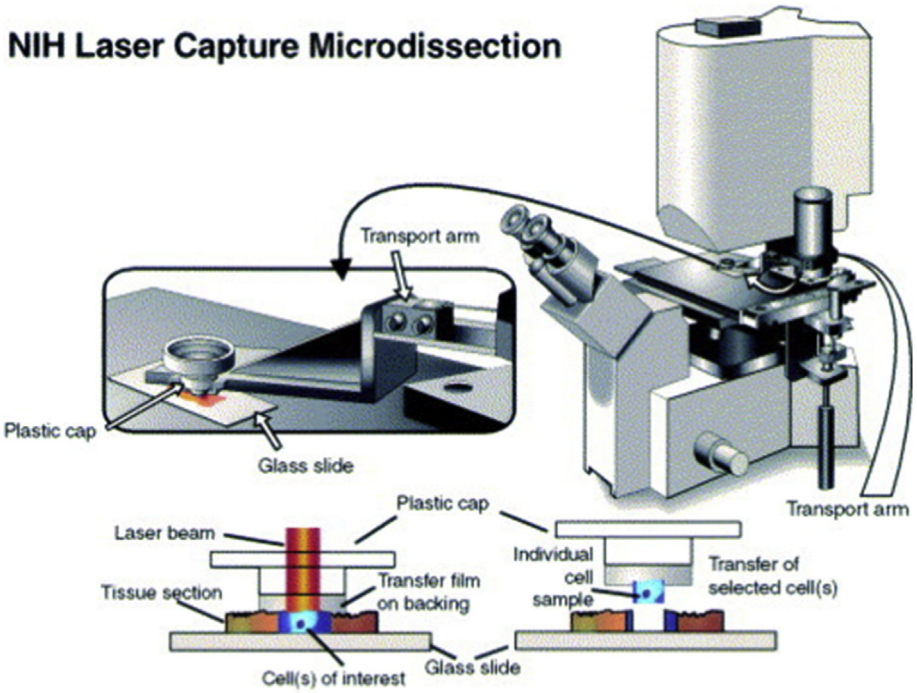


Fig. 8. Schematic of an LCM. Cells of interest are cut and removed from the substrate via laser manipulation. Reprinted with permission from Ref. [64].

3.6. Hydrodynamic sorting

Hydrodynamic sorting is a microfluidic separation technique that, for sexual assault samples, leverages the difference in size between sperm and epithelial cells to provide efficient separations. By manipulating the sample and sheath flow rates, as well as microchip architecture and channel branching, sperm cells (~5 μm diameter) can be separated from epithelial cells (~40 μm diameter) based on their flow along different streamlines. This has been applied by Liu et al., who employed a pinched-expanded channel design to capture sperm cells from mock SAECKs [76] (Fig. 10). This architecture forces cells into alignment against the sidewall, with the smaller cells (sperm) stacking closer to the edge. When the channel rapidly expands, differences in alignment due to cell radius are amplified, resulting in segmentation of sperm and epithelial cells. By performing the separation under laminar flow, cells remain in their streamlines and can be collected into distinct fractions. This sorting method was evaluated using mock sexual assault samples, each containing 50,000 female epithelial cells and 15,000 male sperm cells in a 50 μL sample. STR analysis and flow cytometry indicated a sperm cell recovery of 41%, with greater than 95% purity of male DNA in the sperm fractions. While excellent purity, the capture efficiency is relatively low given the effectiveness of cell separations in microfluidic systems. While the authors do not report on samples containing a low number of sperm cells, it is reasonable to assume that it may be challenging to generate single-source male profiles from such samples. However, this approach does have several notable benefits. The microdevice is simple to use, easy to fabricate, and there is no required differential lysis step before sperm cell capture. With further development, this approach could provide a simple, automated alternative to conventional DE.

A similar passive approach to sperm cell capture was demonstrated by Horsman et al., using sedimentation of epithelial cells and low flow rates to separate sperm cells from mixed samples [77]. When a mock sample composed of intact female epithelial cells and sperm cells was introduced to the microfabricated device, epithelial cells collected faster at the bottom of the sample reservoir. This occurred based on the faster sedimentation rate of epithelial cells as compared to sperm cells, based on differences in their size and specific gravity. In addition, it was hypothesized that adsorption of epithelial cells to the microchannel prevented dispersion throughout the microchip. When a low flow rate was applied,

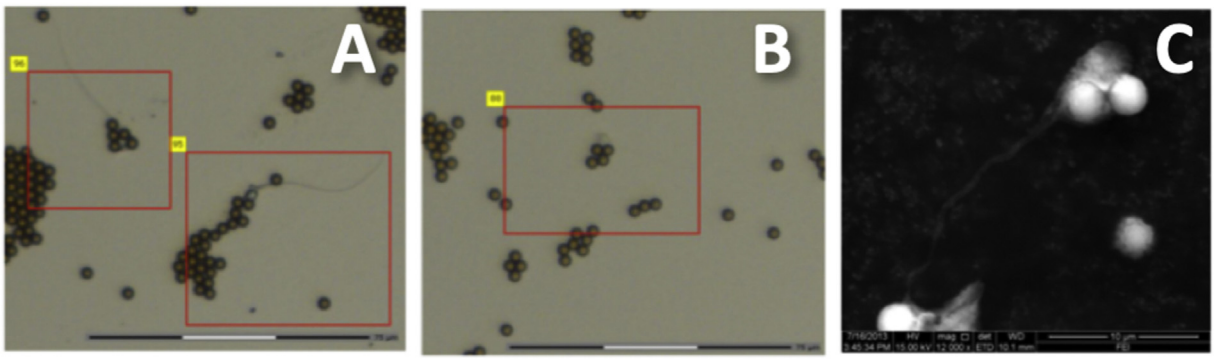


Fig. 9. Sperm cells captured by anti-PH-20 immunogenetic beads. (A) Optical microscopy images for IMB bound to the head of intact sperm cells.(B) Optical microscopy image for IMB bound to the sperm with midpiece and tail. (C) Scanning electron microscopy image for IMB bound to the head sperm cell. Reprinted with permission from Ref. [71].

sperm cells were selectively removed from the reservoir, resulting in an STR profile that was predominantly male, with some female DNA present.

3.7. Acoustic differential extraction

Acoustic forces have previously been leveraged for particle separations throughout microfluidics, material science, molecular diagnostics, and biomedical research [78–83]. Acoustic differential extraction (ADE) removes the manual centrifugation, washing, and enrichment steps associated with conventional DE, and instead, applies microfluidic acoustic separation principles to capture sperm cells from sexual assault samples. ADE separates particles based on their size, density, and compressibility using forces generated by acoustic waves. This is accomplished by balancing acoustic and viscous forces, creating a 'cutoff size' at which only larger particles are trapped. Due to the relatively uniform size, shape, and density of sperm cells [84], reproducible trapping is possible by tuning the flow rate, and importantly, the applied acoustic forces. The most significant force involved is the primary radiation force (PRF), which causes particles to migrate either to the 'nodes' or 'antinodes' of a standing wave, where they are packed together and held in place [85]. The PRF (Equation (1)) for a microfluidic system depends on the acoustic pressure amplitude (ρ_0), volume of the particle (V_c), compressibility of the medium (β_w), wavelength of the standing wave (λ), acoustic contrast factor (φ), wave number (k) and the distance from a pressure node (x); particle density (ρ_c) and compressibility (β_c) compared to the medium density (ρ_w) and compressibility (β_w).

$$F_r = -\left(\frac{\pi\rho_0^2 V_c \beta_w}{2\lambda}\right) \cdot \varphi(\beta,\rho)\sin(2kx) \quad where \quad \varphi(\beta,\rho) = \frac{5\rho_c - 2\rho_w}{2\rho_c + \rho_w} - \frac{\beta_c}{\beta_w} \tag{1}$$

Keeping in mind that many applications of acoustophoresis are in the microfluidic realm, the applied wavelength (λ) must be such that it 'fits' in the sub-millimeter scale resonant chambers [86], with channel height corresponding to a half wavelength of the transducer frequency. This requires acoustic frequencies on the order of 1.5 MHz or greater, proportional to the chamber height of the microdevice. This acoustic contrast factor is important, as it defines whether particle aggregation is at the node or antinode. For ADE, with samples composed of sperm cells surrounded by lysed non-sperm cells, the acoustic contrast factor is positive, and thus, the cells migrate to the nodes of the standing wave. Migration to the node is counteracted by the drag force, which pulls particles along with the bulk fluid flow. As long as the combined acoustic radiation forces are stronger than the drag force acting on the sperm cells, they will remain trapped in the acoustic nodes, while smaller particles (free DNA, cellular debris) are excluded from acoustic trapping. This creates a threshold, where only larger particles (i.e., sperm cells) will be trapped, and smaller particles (DNA from non-sperm cells, debris) will be washed away. Fig. 11 illustrates this selective trapping, as larger particles can be retained under a constant flow rate and application of a standing acoustic wave. A microfluidic platform has been developed for exploiting these acoustic forces, for specifically trapping sperm in lysate.

The primary developments on ADE occurred between 2006 [87] and 2009 [25], where proof-of-principle work demonstrated that intact sperm cells could be captured using ultrasonic standing waves in a glass microchip. By flowing a solution containing sperm cells and non-sperm DNA through a standing acoustic wave, sperm cells were retained and then diverted to a collection reservoir (Fig. 12, reprinted with permission from Ref. [25]). This showed a 16-fold enrichment of sperm cell concentration, and obtained enriched male fractions from mock samples of epithelial cell lysate. While an interesting and potentially important development, this early application of ADE was demonstrated with purified cell lysate (not intact epithelial cells mixed with sperm), and thus was not applicable for the majority of SAECKs. Since that time, ADE has been tested on a wide range of mock mixtures of intact female buccal cells, vaginal cells, sperm cells, along with blood and bacteria. Clean sperm and non-sperm fractions were obtained from mock samples containing an equal number of sperm and epithelial cells, as well as samples with contaminating cells (yeast/bacteria), and samples consisting of a 40:1 ratio of female:male cells [88,89]. Fig. 13 depicts the 'pre-trapping' STR profile from a 20:1 ratio of female:male cells and the 'post-trapping' STR profiles from cells from the non-sperm (R5) and the sperm (R4) fractions [88]. The resulting STR profiles from the 20:1 female:male mixture indicates successful separation of contributing sperm and non-sperm cells.

4. Concluding statements

In the United States and other countries, legislation at both local and national levels has changed how rapidly sexual assault cases are reported, yet the methods for processing DNA samples have remained largely unchanged. This could be due to pragmatic issues (generated data subject to legal scrutiny in the courts), or a lack of substantial improvements in the fundamental analytical processes associated with DNA analysis of sexual assault evidence. DE is the first analytical process in a series of steps that culminate in a

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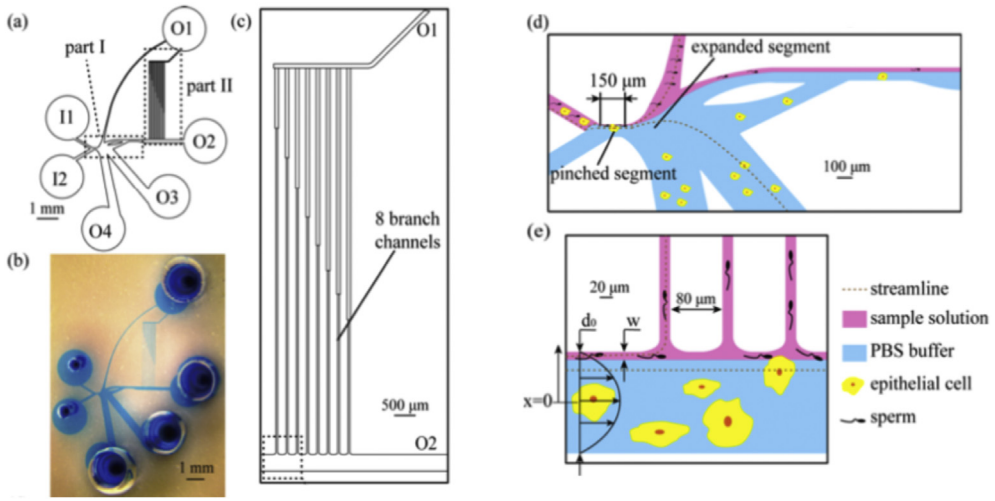


Fig. 10. Microdevice used to separate sperm from EPs, (A) Schematic illustration of the microchannel design, (B) top view of the microdevice, (C) detailed illustration of part II shown in (A), (D) detailed illustration of part I shown in (A), and (E) detailed illustration of the branch points shown in (C). I1, I2,O1, O2, O3, and O4 indicate inlet.1, inlet 2, outlet 1, outlet 2, outlet 3, and outlet 4, respectively, Reprinted with permission from Ref. [76].

genetic profile that can implicate or exclude a male suspect. While 'extraction' is embedded in its name, it is, fundamentally, a separation technique – separate the sperm cells in the sample from the non-sperm fraction of the sample. As a 'differential chemical lysis' method, the elegance lies in its simplicity – effectively lyse all cells except sperm cells, then expose the lysate to centrifugal force, where the highest density components that remain (sperm cells) are separated from the lysate by pelleting. Certainly not a new approach, as the mechanism harkens back to 1924 when Theodor Svedberg was the first to exploit differences in density to separate the macromolecular components of lysed cells by ultracentrifugation [90]. With any approach to DE that is employed, the expectation is that the product of that analytical separation is a sperm fraction that may or may not be completely devoid of DNA from cells other than sperm, and is suitable for mixture deconvolution. In practice, obtaining a profile with a mixture ratio of approximately 4:1 is sufficient to isolate the genotypes of the two contributing donors.

The conventional chemical DE process, while remarkably effective, remains a time-consuming, manual or partially-automated task that is not amenable to complete automation in its current form. With forensic laboratories experiencing an increased demand for the analysis of sexual assault evidence and facing a growing backlog of over 200,000 cases in the U.S. alone [29–33,91], alternative approaches to conventional DE have been deemed worthy of exploration. We posit that any analytical method considered for implementation in a forensic laboratory will be measured by the effectiveness of the separation of the sperm and non-sperm fraction in mixed samples. Additionally, the method should ideally be amenable to automation, inherently rapid and reproducible within and between laboratories.

Some additional, but less-proven, methods are currently in development, but were not described in this review. Capillary zone electrophoresis may be feasible for directly capturing sperm cells from SAECKs, using the different electrophoretic mobilities of sperm and epithelial cells to generate male and female fractions. Another approach is to perform selective PCR from non-separated samples, only amplifying male DNA via Y-STRs; however, Y-STRs are inherited along the paternal line and are therefore not individualizing. In addition, few countries have Y-STR databases available for comparison. Like the methods described, however, neither of these approaches fully address the forensic issues plaguing conventional DE.

While this review is not a comprehensive compilation of DE methods, all of the approaches discussed offer some form of advantage(s) over conventional DE, but none fully-addressing all of the issues limiting the conventional protocol; namely *speed*, *automation*, *throughput* and the *efficient separation* of sperm cells. Some methods are more automated and integrated than DE, e.g., the oocyte capture ligand approach from Inci et al. [62] but lack speed, while others, e.g., the Erase Sperm Isolation Kit, are associated with fewer manual steps. Others, like LCM or the DEPArray™, offer the unique ability to salvage sperm cells from samples with a low total number of cells, but are limited to a low sample throughput. ADE can potentially offer speed, but lacks throughput in its current form. In other words, all of the existing or up-and-coming technologies that could potentially supplant DE bring advantage but forfeit at least one of the analytical requirements. That said, it is noteworthy that many of these amenable to speed and automation are still 'in development', with the possibility that new advances will evolve the process to meet the demands of efficiency and throughput. If we view the requirement of *speed*, *automation*, *throughput* and *efficient separation* as an 'unmet need' for DE in the forensic sector, then this review provides a lens through which we view the creativity of researchers, both basic science and practitioners, who are attempting to develop new technologies that hold the promise to effectively address all of the limitations of current DE methods.

5. Future vision

In its current form, the conventional DE method is difficult to replace with a novel technique because it is:1) well-suited to contemporary downstream analytical methods, and 2) generally effective when measured by the three analytical requirements of a successful DE. Despite its success as an analytical technique, it is cumbersome, hampered by DNA extraction recovery limitations that increase the likelihood for DNA loss, and is difficult to implement with complete automation. That is, it works in most cases, but when the number of sperm are small, the process may fail due to sperm loss in the cell separation steps or sperm DNA loss in the DNA extraction process. It may be instructive to take a 'rearview mirror' approach on how fields outside of forensics have solved or are solving comparable challenges; these can be broadly categorized as *microfluidic* or *chemical*. Focusing solely on approaches that do not require fluorescent labeling, so as to minimize expense of both the instrument and

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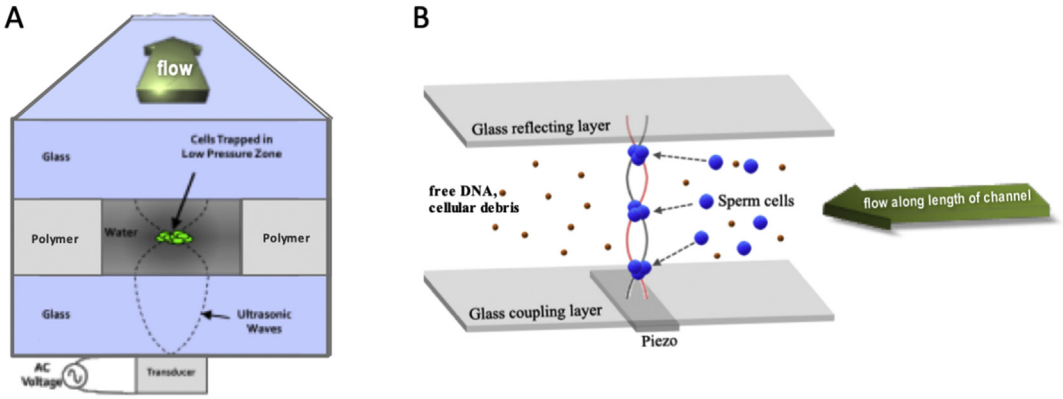


Fig. 11. Acoustic trapping of particles in a channel capped above and bellow by a layer that effectively reflects acoustic waves.(A) Longitudinal view down the channel of a glass-PDMS-glass microchip showing the positioning of the piezoelectric transducer (piezo) and the trapping of particles at a single node. (B) A different view (side) of a trapping chamber where fluid flow is right-to-left and the chamber dimensions conducive to a three-node trapping zone. A piezo vibrates at frequencies in the megahertz range and produces a standing acoustic wave at the resonant frequency of the chamber. When subjected to the standing acoustic wave, particles in solution experience acoustic radiation forces that draw them to the pressure nod(s)e. Larger particles experience a stronger force, allowing for selective trapping based on particle size.

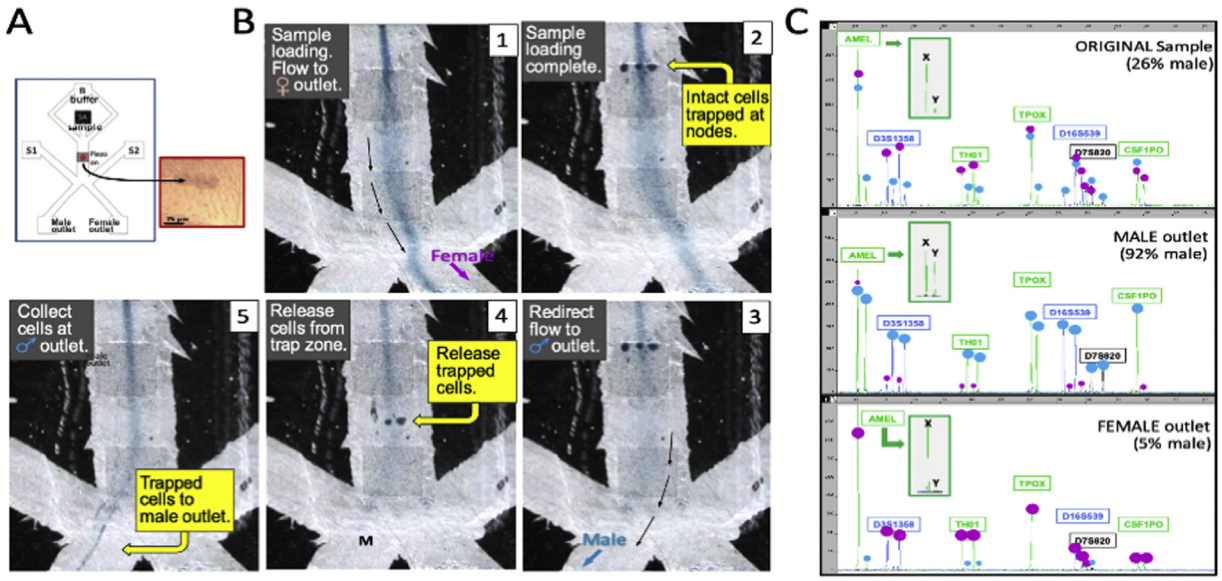


Fig. 12. Fluidic design of the initial ADE chip. (A) Schematic of the ADE design from Norris et al. depicting the buffer inlet (B), sample inlet (SA),piezo, buffer flow control inlets (S1 and S2), and the male and female outlets. The flow ratio of inlets S1 and S2 directs the flow either right to the female outlet or left to the male outlet. (B) 1. Sample loading: sample is infused through the sample the sample inlet and hydrodynamically focused by buffer flow to the female outlet. 2. Intact cells are trapped above the piezo upon activation while lysate components are unretained. 3. Wash: as sample infusion is terminated, the focusing buffer perfuses the trapped cells and removes the residual lysate from the microchannel. Flow is redirected to the male outlet. 4. Release: upon termination of the standing wave, cells are released and diverted to the male outlet. 5. Collection: cells trapped in the male outlet are collected. (C) Top: STR profile of the original mock sample containing a mixture of female epithelial cells and male sperm cells; (peaks representing alleles from sperm DNA and epithelial cell DNA are represented by blue and pink dots). Without ADE, ~26% of DNA is contributed by the male, with allele sharing at amelogenin, TPOX, D16S539, D7S820, and CSF1P0. Middle: STR profile from cells recovered from the male outlet following ADE protocol, of which ~92% of the DNA mixture is male. Lower: STR profile From cells recovered from the female outlet following ADE protocol, of which ~5% of the DNA mixture is male. Reprinted with permission from Ref. [25], copyright (2009) American Chemical Society. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

the consumable, the following touches on several areas that could potentially impact sperm cell isolation in DE-like processes.

The *microfluidic* approaches already discussed in this review incorporated dielectrophoresis, acoustics or hydrodynamics as part of systems that bring a novel tack to DE. One other area with interesting potential is inertial microfluidics. This is a phenomenon where cell sorting is accomplished with microchannel architectures that are spiral or serpentine in design. It exploits the velocity of a fluid medium under laminar flow to force particles of different sizes to different locations across the width of the channel, with the larger particles focused closer to the outer wall. Di Carlo et al. utilized serpentine architecture for the size-based separation of cells, with the clear-cut advantage being volumetric flow rates on the order of 1.5 mL/min without ancillary (sheath) flow [92]. As such, these systems can be high-throughput, require simple pressure-driven flow and, due to the absence of valves, has minimal opportunity for device failure. Another involves the use of ultrasonic waves in the 'acoustophoretic' mode. Unlike the ADE system described earlier, an orthogonal standing acoustic wave is deployed here (across width of the channel), and does not result in particle 'trapping' in a flow stream but, instead, the acoustics augment the 'positioning' of particles across the channel in a size-dependent manner. The Laurell-Nilsson group has been particularly nimble in this space, showing the high throughput capabilities by isolating extremely low abundance cancer cells present in whole blood (1 in a billion cells), with flow rates as fast as 100 μL per min, with

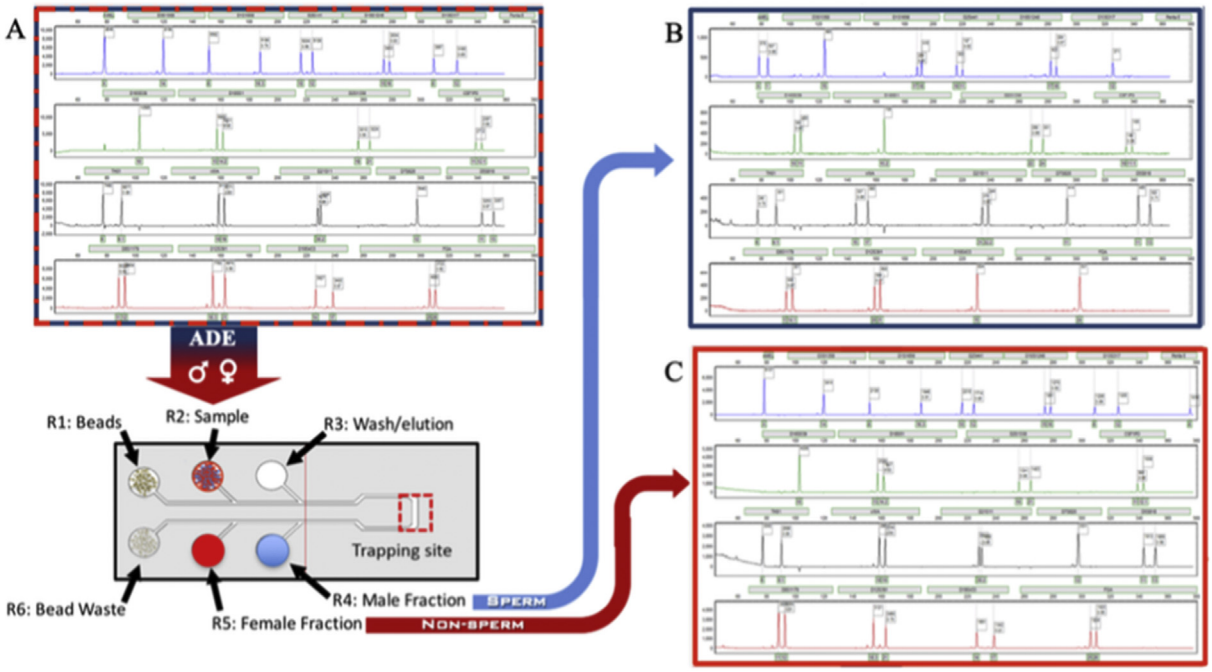


Fig. 13. Glass microchip and ADE of a 20:1 ratio of vaginal epithelial cells:sperm cells. PCR and electrophoretic analysis of the male and female fractions following ADE (acoustic differential extraction) was carried out with an 18-plex STR kit. (A) The glass-PDMS-glass microchip containing a trapping channel that is 1. 1 mm wide, 0.285 mm tall, and 3 mm long. The microchip contains inlet for beads, sample and wash buffer and outlet waste, and male and female fractions. (B) STR profile of the 20:1 female:male mixture from the sample inlet. Male contribution to the sample is minute and not visible at some lici. (C) 'Post-trapping' STR profile sourced from the male collection outlet with minimally visible female DNA contribution. (D) 'Post-trapping' STR profile sourced from the female collection outlet with minimally visible male DNA contribution.

milliliter volumes processed in less than an hour, to enrich cancer cells from whole blood. [93] A final, and very different, microscale approach has been described by Seashols-Williams and coworkers involving the use of 'optical tweezers' [94]. This involves forming an 'optical trap' by directing an optically-focused 1064 nm CW laser beam on sample spread across a microscope slide. When the focal point is placed close to cells ranging from in size from 1 to 100 μM, the cell is 'trapped', and either moving the laser beam while the microscope slide remains stationary, or moving the slide (via translating microscope stage) with a beam is static, allows for controlled movement of a single cell. They show that obtaining sorted fractions of 95–100% sperm cell purity is possible, and with a sperm cell recovery of greater that 90%. The efficiency of this separation process is impressive, but with a per sample time of 20–30 min, it is not currently amenable to high throughput.

The *chemical* approaches involve either enzyme- or affinity-based processes. On the enzymatic side, the selective degradation of residual female DNA, post enzymatic epithelial cell lysis (usually done with Proteinase K & SDS) and centrifugation of sperm is one component of the DE method that is rapid, amenable to automation, and eliminates the need for successive washes and tube transfers. Additionally, the DE product is compatible with the current analytical systems and kit chemistries associated with forensic DNA typing at present. Presently, several commercial 'nuclease-only' products have been developed that produce male STR profiles of similar quality to those generated with the conventional DE method [55–61]. In fact, a solely enzymatic approach is possible through the use of direct lysis (and direct to PCR) capable enzymes such as ForensicGEM and Acrosolv from MicroGEM.[2] These enzymes, used in sequence with a compatible nuclease, can produce separate PCR ready lysates of epithelial cells and sperm. Thus, given the current infrastructure of DNA analysis, the complete automation of a robust selective degradation method to replace conventional DE is conceivable. On the affinity-based side, there are non-traditional immune-capture/sorting approaches that have potential in the DE space. In a 2016 report, Sakar et al. described a number of affinity-based cell isolation methods but emphasized the limitations of binary 'bind-elute' approaches, particularly with the isolation of multiple low-abundance cell types. They describe a rapid multiplexed, affinity-based cell isolation method, combining a size-coded mixture of multiple antibody-bound microbeads but, uniquely, is coupled with inertial microfluidics for particle sorting. They provide demonstrable proof-of-feasibility with the capture of rare HIV-specific cells ($CD4^+$) from a crude matrix like whole blood [95]. In a non-antibody affinity-based approach, Zhang et al. exploit surface nanotechnology in the form of electrospun titanium oxide (TiO_2) nanofibers (TiNFs) with dimensions similar to cellular components [96]. They show that controlling the diameter of the electrospinning process and the packing density of the TiNFs, that nanotextured surfaces can be devised with high specificity for isolating rare circulating tumor cells (CTCs) from whole blood.

Finally, forensic samples present an additional challenge to microfluidics that is not necessarily a problem in the macroworld – sample volume. Microfluidic systems routinely accept sample volumes on the order of tens of microliters, which may be adequate for some forensic scenarios, but unacceptably low with others. This is the macro-to-micro interface problem that the microfluidics sector has been addressing since inception of the micro-total analysis system concept conceived by Manz, Graber and Widmer in 1990 [97]. Tens of microliters of sample from an SAEK may be adequate for some microfluidic DE methodologies, but there are scenarios that yield milliliters of sample for DE processing, e.g., the wash process involved with removing all of the semen from evidence recovered at a crime scene. While it is important to keep the macro-to-micro interface

[2] A full description of reagents is available upon request at the corresponding author.

problem in mind when developing new approaches for DE, it is clear that developments in the microfluidic arena have found ways to solve this. Examples include those described above for the recovery of rare cells in whole blood (CTC) [85,93,98]. This group was forced to solve this problem early on (2004) where they exploited acoustics as a real-time filter to physically isolate lipids from blood [99]. While the microfluidic device was fabricated in silicon (not glass or polymeric material) and roughly the size of a microscope slide, it allowed for a high volumetric flow rate processing of several liters of blood over a period of hours. Hence, incongruence between the sample volume and 'microfluidic' should not be viewed as an insurmountable barrier.

In summary, due to the variation in types of evidence samples and amount of cellular material deposited on them, it likely that there will be no single, all-encompassing technology provides a universal solution. That said, it is critical that pursuit of technology for this (or any other) forensic application continues to involve 'thinking outside the box', and doing so in close collaboration with the forensic scientists, who may not have resources for R&D, but unequivocally know exactly what is needed to maximally impact workflow in forensic labs.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Charles Clark received his PhD in Chemistry from the University of Virginia (Virginia, United States) in 2019, where he used acoustic separations for microfluidic sperm cell capture and developed electrodriven separations of explosive compounds. Charles has collaborated with multiple forensic laboratories in the United States to study and improve the differential extraction process for sexual assault samples. He is currently a Postdoctoral Researcher with the Leiden Academic Center for Drug Research (Leiden, Netherlands), continuing to develop microfluidic separation techniques.



Rachelle Turiello is a Ph.D. student with the Chemistry Department at the University of Virginia. She holds a bachelor's degree from the University of Maryland Baltimore County in Interdisciplinary Studies: Bio-Cultural Hominid Anthropology, and a master's degree from Towson University in Forensic Science. Rachelle has worked as a civilian scientist for multiple police departments in crime scene investigation and forensic DNA analysis. Her research is focused on centrifugal microfluidic devices for human identification.



Dr. Cotton has a Ph.D. in Molecular Biology and Biochemistry from the University of California at Irvine in 1980 as well as 18 years of experience in the forensic DNA applications. She has testified in DNA results in 250 criminal cases in 35 states. In October of 2006, Dr. Cotton joined the faculty at the Boston University School of Medicine as the Director of the M.S. Program in Biomedical Forensic Sciences. Dr. Cotton's research focuses on development of new DNA extraction methods for low template DNA samples. She is a member of Forensic Advisory Committees for Virginia and Massachusetts and is appointed to the NIST Scientific Advisory Committee on Biology and DNA.



James Landers is a Jefferson Scholars Fellow and Commonwealth Professor of Chemistry, Professor of Mechanical Engineering, and an Associate Professor of Pathology at the University of Virginia. He holds both bachelor and PhD degrees in Biochemistry from the University of Guelph in Canada, and post-doctoral fellowships at the Hospital for Sick Children in Toronto, the University of Toronto School of Medicine and the Mayo Clinic. His research program focuses on the development of fully-integrated microfluidic systems for sample-to-result genetic, biochemical and chemical analysis. He has published more than 250 papers, 25 book chapters and founded two start-up companies.

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